Supplement to: Analysis choices and orbital sampling shape a particle-defined South Atlantic Anomaly footprint

Extended methods, full configuration results, data accounting, quality-rule evidence and descriptive magnetic material

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This supplement documents every result family in full. Machine-readable versions of all tables are in `paper/tables/`. Bulky rows are linked, not printed: bootstrap draws (`outputs/channel_assurance/block_draws.json`, `replicates.csv`, `paired_replicates.csv`), selected cells (`outputs/tables/cp4d_temporal_support_selected_cells.csv`) and distinct-day counts (`outputs/tables/cp4d_temporal_support_distinct_days.csv`). Every number printed in the text is generated from the claims register `paper/generated/claims_register.json`, which lists, for each claim, its configuration, input paths with SHA-256, formula, denominator, rounding and caveat. Rounded values are for display; comparisons use unrounded values.

S1 Inputs and product provenance

Files. 155 daily NetCDF files (five spacecraft × 31 days of January 2024) from the NCEI archive path `…/access/l1b/v01r00/2024/<satellite>/poes_<token>_202401DD_proc.nc`. `paper/tables/S1_raw_manifest.csv` lists name, size, SHA-256 and URL for each file, and the reproduction verifies every hash against the recorded Phase 1 manifest before computing. Raw files are not redistributed with the package.

Product level and units. The archive directory is `l1b`, but every file declares `processing_level = "Level 2, calibrated fluxes"`. The NOAA manual describes the processing as turning level-1B sensor data into physical-unit level-2 products (Green, 2013). Variable metadata define `mep_omni_flux_p1..p3` as "MEPED proton differential flux at 25/50/100 MeV omnidirection telescope" in `#/cm2-s-str-MeV`. The manual's Table A-1 (p. 17) prints `#/cm2-s-str`. The OMNI algorithm description (Machol, 2012) (§3.1 p. 10; Table 4 pp. 24–25) defines the outputs as fitted differential fluxes at 25, 50 and 100 MeV in ions/(cm² s sr MeV). We treat the manual's label as an incomplete unit string and report per-MeV differential flux. This reconciles documentation; it does not define a new product. The OMNI fluxes are power-law fits in energy to count rates from four overlapping omnidirectional channels (16–35, 35–250, 70–250, 140–250 MeV), so p1–p3 share inputs and a spectral model.

Fit and IFC flags. ATBD Table 5 (p. 25): fit −1 = no fit performed; 0 = four-region power-law fit; 1 = one-point fit with a single slope set to a default; 2 = two-point single-slope fit. `mep_IFC_on` is documented as 0 (off) / 1 (on); the value −1 that dominates the files is undocumented.

Provider caveat. The file title states that the data have known contamination problems and advises consulting the provider for usage. No specific contamination correction is applied here.

Quality-evidence input. One level-1a companion file from the same archive, `…/access/l1a/v01r00/2024/noaa15/poes_n15_20240119_raw.nc` (SHA-256 `59cd88dd551ac2a18036137124000e1e91c74dd26e29233a57821a062a95626f`, recorded in `paper/inputs/qc_evidence_hashes.json`), holds the OMNI count rates of the day of the excluded record. Its record times are identical to those of the processed file. It is read only to document that record (S7) and never enters a map.

Historical bytes. The 31 NOAA-19 files were cached with recorded hashes before this analysis was finalized. The other 124 were retrieved again. Regional tables rebuilt from them reproduce the saved values exactly, which is value lineage, not recovery of unrecorded historical bytes.

S2 Quality handling and record accounting

Per spacecraft (region lat −70°…20°, lon −100°…20°): regional records before the IFC rule, records dropped with `mep_IFC_on = 1`, records retained with the undocumented −1, p1 values set missing by the validity rule (negative

or fit = −1), p1 values excluded by QC-1, and records remaining. The principal footprint sample counts and descriptive field-strength columns belong to S8.

satellite	regional	IFC=1 dropped	IFC=−1 kept	p1 invalid	QC-1	analysed	L invalid
noaa15	207233	1	194253	25	1	207206	5786
noaa18	208722	0	195676	4	0	208718	3974
noaa19	205153	0	192334	0	0	205153	3914
metop01	227572	0	213350	0	0	227572	5523
metop03	226645	0	212478	0	0	226645	5542

The validity rule sets a value missing only when it is negative (below the documented minimum of 0) or its fit flag is −1. Valid zeros remain in means and medians, and log colour scales cannot display them. No upper cutoff is imposed. Missing flags remain missing; a synthetic regression test covers fill, scale and separate-dimension alignment in the loader. QC-1 (main text Sect. 2.2; evidence in S7) is applied to each daily file in native record order, before region selection and gridding, so neighbours outside the region count as neighbours. Its exclusions are counted separately from the validity rule; the record-level ledger, with the reason for each exclusion, is `outputs/tables/qc1_omni_exclusions.csv`.

S3 Estimands, configuration universe and threshold family

Percentile. c_q is NumPy's default linear-interpolation percentile over the eligible cells' statistic. Selection is $F_i \geq c_q$, so ties at the cutoff are all selected.

Area. $A_i = R^2 \Delta\lambda (\sin\varphi_i^+ - \sin\varphi_i^-)$, with $R = 6371$ km and $\Delta\lambda$ in radians. An independent check uses $2R^2 \Delta\lambda \cos\varphi_i \sin(\Delta\varphi/2)$, which is algebraically identical, and agrees to 10^{-12} relative.

Centroid. Coordinate moment with weights $F_i A_i$ (main-text Eq. 2). Longitude is averaged directly because the region spans −100°…20°. Distances are haversine and are checked against a vector-angle formula.

Configuration universe. The viewer payload lists 360 configurations. Grouping by grid content, statistic and threshold gives 280 distinct states. The NOAA-19 monthly p1 map appears as the threshold family, the p1 member of the channel family, the month member of the time family and the NOAA-19 member of the satellite family (20 states × 4 listings), and `week1` duplicates the cumulative 1–7 January window (20 × 2). `paper/generated/configuration_aliases.json` lists the alias groups. Configuration counts are therefore not counts of independent experiments.

Threshold family, full results (NOAA-19 p1, January; flux cutoff and peak in cm^{-2} s^{-1} sr^{-1} MeV^{-1} area in 10^6 km^2):

grid	statistic	threshold	cutoff	eligible	selected	area_Mkm2	lat	lon	peak
5	mean	top20	5.139	432	87	24.998	-19.743	-55.075	31.462
5	mean	top10	12.564	432	44	12.614	-20.886	-55.479	31.462
5	mean	top5	19.763	432	22	6.309	-21.433	-55.642	31.462
5	mean	top2	26.29	432	9	2.574	-21.938	-55.25	31.462
5	mean	top1	28.69	432	5	1.416	-23.425	-56.425	31.462
5	median	top20	4.88	432	87	24.984	-19.737	-54.845	31.516
5	median	top10	12.767	432	44	12.614	-20.858	-55.459	31.516
5	median	top5	19.848	432	22	6.309	-21.4	-55.786	31.516
5	median	top2	26.365	432	9	2.574	-21.912	-55.243	31.516
5	median	top1	28.776	432	5	1.416	-23.427	-55.538	31.516
2	mean	top20	4.772	2685	537	24.703	-19.746	-54.923	32.401
2	mean	top10	12.914	2685	269	12.371	-20.635	-55.612	32.401
2	mean	top5	20.803	2685	135	6.183	-21.617	-55.474	32.401
2	mean	top2	27.127	2685	54	2.46	-22.659	-55.681	32.401
2	mean	top1	29.568	2685	27	1.228	-22.961	-56.352	32.401
2	median	top20	4.768	2685	537	24.697	-19.715	-54.979	32.451
2	median	top10	13.128	2685	269	12.371	-20.604	-55.713	32.451
2	median	top5	20.785	2685	135	6.174	-21.76	-56.317	32.451
2	median	top2	26.875	2685	56	2.563	-22.062	-55.42	32.451
2	median	top1	29.197	2685	32	1.457	-22.765	-55.843	32.451

Successive 5° mean centroid steps are 134, 63, 69 and 205 km. The selected-cell ratios top 20% / top 1% are 17.4 (5° mean), 17.4 (5° median), 19.9 (2° mean) and 16.8 (2° median); the nominal ratio is 20.

S4 Channel family

Footprint results for p1, p2 and p3 across all 20 settings (channel absolute values differ by construction and are not compared):

ch	grid	statistic	threshold	cutoff	selected	area_Mkm2	lat	lon
p1	5	mean	top20	5.139	87	24.998	-19.743	-55.075
p1	5	mean	top10	12.564	44	12.614	-20.886	-55.479
p1	5	mean	top5	19.763	22	6.309	-21.433	-55.642
p1	5	mean	top2	26.29	9	2.574	-21.938	-55.25
p1	5	mean	top1	28.69	5	1.416	-23.425	-56.425
p1	5	median	top20	4.88	87	24.984	-19.737	-54.845
p1	5	median	top10	12.767	44	12.614	-20.858	-55.459
p1	5	median	top5	19.848	22	6.309	-21.4	-55.786
p1	5	median	top2	26.365	9	2.574	-21.912	-55.243
p1	5	median	top1	28.776	5	1.416	-23.427	-55.538
p1	2	mean	top20	4.772	537	24.703	-19.746	-54.923
p1	2	mean	top10	12.914	269	12.371	-20.635	-55.612
p1	2	mean	top5	20.803	135	6.183	-21.617	-55.474
p1	2	mean	top2	27.127	54	2.46	-22.659	-55.681
p1	2	mean	top1	29.568	27	1.228	-22.961	-56.352
p1	2	median	top20	4.768	537	24.697	-19.715	-54.979
p1	2	median	top10	13.128	269	12.371	-20.604	-55.713
p1	2	median	top5	20.785	135	6.174	-21.76	-56.317
p1	2	median	top2	26.875	56	2.563	-22.062	-55.42
p1	2	median	top1	29.197	32	1.457	-22.765	-55.843
p2	5	mean	top20	3.43	87	24.85	-20.84	-54.171
p2	5	mean	top10	8.278	44	12.541	-21.777	-54.807
p2	5	mean	top5	13.324	22	6.252	-22.62	-55.786
p2	5	mean	top2	17.898	9	2.533	-24.05	-56.841
p2	5	mean	top1	19.472	5	1.405	-24.391	-58.36
p2	5	median	top20	3.379	87	24.85	-20.827	-54.181
p2	5	median	top10	8.237	44	12.541	-21.771	-54.808
p2	5	median	top5	13.111	22	6.252	-22.615	-55.8
p2	5	median	top2	17.685	9	2.533	-24.031	-56.829
p2	5	median	top1	19.323	5	1.405	-24.394	-58.337
p2	2	mean	top20	3.413	537	24.513	-20.876	-54.205
p2	2	mean	top10	8.819	269	12.27	-21.709	-54.979
p2	2	mean	top5	13.842	135	6.142	-22.555	-55.314
p2	2	mean	top2	18.063	54	2.448	-23.308	-56.659
p2	2	mean	top1	20.018	27	1.223	-23.55	-55.927
p2	2	median	top20	3.381	537	24.502	-20.896	-54.245
p2	2	median	top10	8.822	269	12.267	-21.735	-55.11
p2	2	median	top5	13.919	135	6.145	-22.485	-55.322
p2	2	median	top2	17.694	54	2.446	-23.401	-56.294
p2	2	median	top1	20.134	27	1.227	-23.104	-56.091
p3	5	mean	top20	2.182	87	24.767	-21.142	-53.5
p3	5	mean	top10	5.257	44	12.531	-21.947	-53.795
p3	5	mean	top5	8.147	22	6.218	-23.166	-55.147
p3	5	mean	top2	11.174	9	2.533	-24.087	-56.785
p3	5	mean	top1	12.298	5	1.405	-24.404	-55.362
p3	5	median	top20	2.097	87	24.817	-21.042	-53.582
p3	5	median	top10	5.133	44	12.531	-21.94	-53.824
p3	5	median	top5	7.962	22	6.252	-22.669	-55.497
p3	5	median	top2	10.909	9	2.533	-24.106	-56.871
p3	5	median	top1	11.939	5	1.405	-24.403	-59.173
p3	2	mean	top20	2.161	537	24.554	-20.924	-53.638
p3	2	mean	top10	5.472	269	12.245	-22.054	-54.25
p3	2	mean	top5	8.697	135	6.126	-22.871	-54.534
p3	2	mean	top2	11.3	54	2.442	-23.606	-55.497
p3	2	mean	top1	12.47	27	1.223	-23.503	-55.387
p3	2	median	top20	2.15	537	24.56	-20.908	-53.631
p3	2	median	top10	5.515	269	12.251	-22.001	-54.357
p3	2	median	top5	8.677	135	6.121	-22.992	-54.426
p3	2	median	top2	11.25	54	2.438	-23.707	-55.723
p3	2	median	top1	12.135	27	1.219	-23.8	-55.914

Pairwise morphology for the four grid/statistic settings. Modes: `original_percentile` (each channel's own eligible support), `common_percentile` (top 10% recomputed on the three-channel common support; identical here because the supports coincide), and `common_continuous` (whole-field normalized moments; "overlap" is Eq. 4 of the main text; "jaccard" is the area-weighted Jaccard of the selected sets, trivially 1 for the whole field). Signed differences are second channel minus first:

grid	statistic	pair	mode	dlat	dlon	km	overlap	jaccard
5	mean	p1/p2	original_percentile	-0.891	0.673	121.096		0.913
5	mean	p1/p2	common_percentile	-0.891	0.673	121.096		0.913
5	mean	p1/p2	common_continuous	-1.053	0.562	131.008	0.947	1.0
5	mean	p1/p3	original_percentile	-1.06	1.684	210.442		0.872
5	mean	p1/p3	common_percentile	-1.06	1.684	210.442		0.872
5	mean	p1/p3	common_continuous	-0.933	0.945	143.191	0.933	1.0
5	mean	p2/p3	original_percentile	-0.169	1.012	106.07		0.955
5	mean	p2/p3	common_percentile	-0.169	1.012	106.07		0.955
5	mean	p2/p3	common_continuous	0.12	0.382	41.965	0.972	1.0
5	median	p1/p2	original_percentile	-0.913	0.65	121.797		0.913
5	median	p1/p2	common_percentile	-0.913	0.65	121.797		0.913
5	median	p1/p2	common_continuous	-1.067	0.6	134.151	0.945	1.0
5	median	p1/p3	original_percentile	-1.082	1.635	207.616		0.872
5	median	p1/p3	common_percentile	-1.082	1.635	207.616		0.872
5	median	p1/p3	common_continuous	-0.91	0.931	140.304	0.931	1.0
5	median	p2/p3	original_percentile	-0.169	0.985	103.332		0.955
5	median	p2/p3	common_percentile	-0.169	0.985	103.332		0.955
5	median	p2/p3	common_continuous	0.158	0.331	38.689	0.969	1.0
2	mean	p1/p2	original_percentile	-1.075	0.632	136.303		0.883
2	mean	p1/p2	common_percentile	-1.075	0.632	136.303		0.883
2	mean	p1/p2	common_continuous	-1.057	0.564	131.505	0.944	1.0
2	mean	p1/p3	original_percentile	-1.42	1.362	211.675		0.83
2	mean	p1/p3	common_percentile	-1.42	1.362	211.675		0.83
2	mean	p1/p3	common_continuous	-0.935	0.94	143.019	0.929	1.0
2	mean	p2/p3	original_percentile	-0.345	0.73	84.48		0.921
2	mean	p2/p3	common_percentile	-0.345	0.73	84.48		0.921
2	mean	p2/p3	common_continuous	0.123	0.376	41.468	0.969	1.0
2	median	p1/p2	original_percentile	-1.131	0.603	140.415		0.883
2	median	p1/p2	common_percentile	-1.131	0.603	140.415		0.883
2	median	p1/p2	common_continuous	-1.136	0.711	146.473	0.936	1.0
2	median	p1/p3	original_percentile	-1.397	1.356	209.406		0.829
2	median	p1/p3	common_percentile	-1.397	1.356	209.406		0.829
2	median	p1/p3	common_continuous	-0.976	1.057	154.864	0.919	1.0
2	median	p2/p3	original_percentile	-0.266	0.753	83.145		0.901
2	median	p2/p3	common_percentile	-0.266	0.753	83.145		0.901
2	median	p2/p3	common_continuous	0.159	0.346	40.154	0.953	1.0

S5 Paired calendar-block resampling

Protocol (frozen before production replicates; `outputs/channel_assurance/protocol.json`, version 1.0). Monthly p1/p2/p3, 5° mean, top 10%, ≥ 30 samples per cell, NumPy linear percentile. For each block length $L \in \{1, 3, 7\}$, 1,000 replicates. Starts are uniform on $0..31 - L$, drawn with replacement, without wrap; draws are concatenated and trimmed to 31 day-slots. The generator is PCG64 with seed 20260929, reset per block length. The same day draws are used for all three channels, and repeated days keep their multiplicity. Missing days would keep empty slots; none occurred, although within-day gaps remain. Each replicate recomputes coverage, percentile, footprint and centroid from daily cell sums and counts. Nine real-data checks confirm that the sufficient-statistic path reproduces explicitly repeated records (cell means, masks, centroids) to 10^{-12}. All 3,000 draws and 9,000 channel-level results were defined; none was redrawn.

Quantiles (pandas linear; signed second – first; degrees or km; denominator 1,000 defined replicates each):

block_length	pair	metric	defined	q025	q50	q975
1	p2−p1	delta_lat	1000	-1.333	-0.99	-0.687
1	p2−p1	delta_lon	1000	0.062	0.647	1.271
1	p2−p1	distance_km	1000	85.393	130.489	181.884
1	p3−p1	delta_lat	1000	-1.513	-1.149	-0.774
1	p3−p1	delta_lon	1000	0.576	1.429	2.171
1	p3−p1	distance_km	1000	130.682	197.862	262.584
1	p3−p2	delta_lat	1000	-0.3	-0.121	0.081
1	p3−p2	delta_lon	1000	0.28	0.947	1.282
1	p3−p2	distance_km	1000	31.657	99.334	134.432
3	p2−p1	delta_lat	1000	-1.335	-0.992	-0.778
3	p2−p1	delta_lon	1000	0.023	0.677	1.299
3	p2−p1	distance_km	1000	101.352	135.72	182.286
3	p3−p1	delta_lat	1000	-1.464	-1.179	-0.869
3	p3−p1	delta_lon	1000	0.557	1.454	2.209
3	p3−p1	distance_km	1000	152.614	198.751	263.116
3	p3−p2	delta_lat	1000	-0.302	-0.162	0.076
3	p3−p2	delta_lon	1000	0.284	0.902	1.274

block_length	pair	metric	defined	q025	q50	q975
3	p3–p2	distance_km	1000	32.122	96.445	131.972
7	p2–p1	delta_lat	1000	-1.243	-0.958	-0.736
7	p2–p1	delta_lon	1000	0.191	0.618	1.152
7	p2–p1	distance_km	1000	98.432	123.666	169.324
7	p3–p1	delta_lat	1000	-1.306	-1.089	-0.816
7	p3–p1	delta_lon	1000	0.8	1.454	2.065
7	p3–p1	distance_km	1000	151.634	193.268	248.681
7	p3–p2	delta_lat	1000	-0.29	-0.162	0.035
7	p3–p2	delta_lon	1000	0.311	0.971	1.07
7	p3–p2	distance_km	1000	34.236	101.627	111.437

Dependence diagnostics (daily regional-mean p1–p3; descriptive only):

channel	lag_days	lag_correlation	first_half_mean	second_half_mean
p1	1	0.132	3.487	3.273
p1	3	0.097	3.487	3.273
p1	7	0.206	3.487	3.273
p2	1	0.138	2.357	2.211
p2	3	0.101	2.357	2.211
p2	7	0.104	2.357	2.211
p3	1	0.166	1.463	1.384
p3	3	0.089	1.463	1.384
p3	7	0.09	1.463	1.384

Caveats. Thirty-one days cannot establish stationarity or a long-run dependence structure (Künsch, 1989; Synowiecki, 2007). Seven-day blocks have only 25 possible starts and need five draws, the last trimmed to three days. Dates near the month's ends are selected less often when $L > 1$. Joins between blocks break dependence, and 1-day blocks remove cross-day dependence altogether. The thresholded centroid is nonlinear and coverage-dependent. The channels are fitted outputs of shared counts. The quantiles are marginal (they do not form a joint region), are conditional on the month observed, and are not calibrated confidence intervals for a physical position. A positive distance quantile is not significance.

S6 Temporal coverage versus conditional morphology

Definitions (main text §2.6). For each window W and the January reference M at the same grid, statistic and threshold: C_W, C_M are the eligible cells with a finite statistic; F_W, F_M the original footprints; "J_end" $= J(F_W, F_M)$ (area-weighted); "O_M" $= A(F_M \cap C_W)/A(F_M)$; "S" $= |C_W \cap C_M|$; "J_shared" and "D_shared" compare footprints recomputed with each map's own percentile over $C_W \cap C_M$; "R" is the selected-area ratio window/month. Undefined cases would be reported with reasons; there were 0 among 180 rows. Recomputed original footprints match the stored time-window table exactly (cell counts, areas, cutoffs, centroid distances), and three rows were recomputed independently with hand-coded percentiles, zone-formula areas and vector-angle distances. `wk 1*` is the same date range as `7 d`, and its rows are identical.

Window coverage (eligible = cells passing the window's sample minimum):

window	grid	days	min_samples	nominal	observed	eligible	eligible_frac
day_2024-01-01	5	1	3	432	206	201	0.465
day_2024-01-01	2	1	3	2700	548	517	0.191
days_2024-01-01_to_03	5	3	3	432	328	326	0.755
days_2024-01-01_to_03	2	3	3	2700	1336	1276	0.473
days_2024-01-01_to_07	5	7	7	432	429	429	0.993
days_2024-01-01_to_07	2	7	7	2700	2298	2097	0.777
days_2024-01-01_to_14	5	14	14	432	432	432	1.0
days_2024-01-01_to_14	2	14	14	2700	2646	2405	0.891
month_2024-01	5	31	30	432	432	432	1.0
month_2024-01	2	31	30	2700	2700	2685	0.994
week1	5	7	7	432	429	429	0.993
week1	2	7	7	2700	2298	2097	0.777
week2	5	7	7	432	428	428	0.991
week2	2	7	7	2700	2333	2136	0.791
week3	5	7	7	432	432	432	1.0
week3	2	7	7	2700	2404	2214	0.82
week4	5	7	7	432	429	427	0.988
week4	2	7	7	2700	2407	2236	0.828

Full diagnostics (all windows except the reference; 2 grids × 2 statistics × 5 thresholds):

window	grid	stat	thr	C_W	C_W area	J_end	O_M	S	J_sh	D_end km	D_sh km	R_end	R_sh
1 d	5	mean	top20	201	0.448	0.378	0.405	201	0.825	216.883	56.613	0.476	1.003
1 d	5	mean	top10	201	0.448	0.329	0.368	201	0.823	289.117	173.992	0.484	1.0
1 d	5	mean	top5	201	0.448	0.224	0.275	201	0.826	214.857	75.991	0.505	1.001
1 d	5	mean	top2	201	0.448	0.396	0.557	201	0.656	166.623	141.505	0.56	1.005
1 d	5	mean	top1	201	0.448	0.334	0.403	201	0.501	372.036	390.066	0.611	1.024
1 d	5	median	top20	201	0.448	0.379	0.405	201	0.825	215.429	64.557	0.476	1.003
1 d	5	median	top10	201	0.448	0.329	0.368	201	0.823	288.676	181.083	0.484	1.0
1 d	5	median	top5	201	0.448	0.224	0.275	201	0.828	147.671	102.292	0.503	0.998
1 d	5	median	top2	201	0.448	0.396	0.557	201	0.656	159.912	138.946	0.56	1.005
1 d	5	median	top1	201	0.448	0.334	0.597	201	0.501	320.091	392.249	0.611	1.024
1 d	2	mean	top20	517	0.184	0.176	0.184	517	0.891	81.637	18.49	0.193	1.0
1 d	2	mean	top10	517	0.184	0.13	0.149	517	0.823	82.62	114.705	0.195	1.0
1 d	2	mean	top5	517	0.184	0.15	0.156	517	0.855	91.748	38.815	0.193	0.995
1 d	2	mean	top2	517	0.184	0.182	0.222	517	0.833	49.53	103.265	0.204	1.006
1 d	2	mean	top1	517	0.184	0.179	0.297	517	0.498	157.194	185.206	0.224	1.009
1 d	2	median	top20	517	0.184	0.172	0.178	517	0.907	83.369	18.99	0.194	1.002
1 d	2	median	top10	517	0.184	0.134	0.141	517	0.854	114.772	119.053	0.195	1.002
1 d	2	median	top5	517	0.184	0.15	0.156	517	0.855	176.833	44.725	0.193	0.995
1 d	2	median	top2	517	0.184	0.196	0.232	517	0.833	30.027	104.573	0.196	1.006
1 d	2	median	top1	517	0.184	0.189	0.313	517	0.498	125.521	202.531	0.189	1.009
3 d	5	mean	top20	326	0.745	0.667	0.718	326	0.911	162.339	73.994	0.765	1.004
3 d	5	mean	top10	326	0.745	0.54	0.615	326	0.829	138.275	14.271	0.754	0.999
3 d	5	mean	top5	326	0.745	0.442	0.543	326	0.79	26.304	161.914	0.771	0.988
3 d	5	mean	top2	326	0.745	0.604	0.783	326	0.751	288.987	158.014	0.775	0.99
3 d	5	mean	top1	326	0.745	0.285	0.605	326	0.332	548.476	418.269	0.819	1.026
3 d	5	median	top20	326	0.745	0.667	0.719	326	0.908	171.228	67.795	0.765	1.0
3 d	5	median	top10	326	0.745	0.51	0.615	326	0.783	212.075	56.93	0.76	1.002
3 d	5	median	top5	326	0.745	0.5	0.591	326	0.79	13.146	165.678	0.771	0.988
3 d	5	median	top2	326	0.745	0.604	0.783	326	0.751	285.151	154.385	0.775	0.99
3 d	5	median	top1	326	0.745	0.285	0.798	326	0.332	484.972	419.927	0.819	1.026
3 d	2	mean	top20	1276	0.458	0.422	0.456	1275	0.91	137.174	37.153	0.48	1.001
3 d	2	mean	top10	1276	0.458	0.336	0.391	1275	0.837	159.837	38.841	0.479	1.001
3 d	2	mean	top5	1276	0.458	0.284	0.327	1275	0.798	112.751	73.067	0.477	1.001
3 d	2	mean	top2	1276	0.458	0.38	0.465	1275	0.791	328.597	55.049	0.484	1.001
3 d	2	mean	top1	1276	0.458	0.291	0.521	1275	0.526	336.505	118.856	0.485	1.0
3 d	2	median	top20	1276	0.458	0.419	0.452	1275	0.904	136.479	35.981	0.48	1.002
3 d	2	median	top10	1276	0.458	0.327	0.387	1275	0.8	136.886	59.675	0.48	1.004
3 d	2	median	top5	1276	0.458	0.291	0.334	1275	0.839	90.281	47.872	0.483	1.015
3 d	2	median	top2	1276	0.458	0.408	0.466	1275	0.827	300.021	62.916	0.482	1.036
3 d	2	median	top1	1276	0.458	0.364	0.564	1275	0.625	251.883	52.952	0.408	1.0
7 d	5	mean	top20	429	0.993	0.945	0.989	429	0.956	30.376	25.866	0.989	1.001
7 d	5	mean	top10	429	0.993	0.887	0.977	429	0.907	38.872	41.534	0.979	1.001
7 d	5	mean	top5	429	0.993	0.829	1.0	429	0.829	73.611	73.611	1.004	1.004
7 d	5	mean	top2	429	0.993	0.795	1.0	429	0.795	64.791	64.791	1.0	1.0
7 d	5	mean	top1	429	0.993	0.661	1.0	429	0.661	150.439	150.439	1.007	1.007
7 d	5	median	top20	429	0.993	0.942	0.989	429	0.953	58.091	35.074	0.99	1.001
7 d	5	median	top10	429	0.993	0.931	0.977	429	0.952	78.778	45.819	0.978	1.001
7 d	5	median	top5	429	0.993	0.829	1.0	429	0.829	83.354	83.354	1.004	1.004
7 d	5	median	top2	429	0.993	0.795	1.0	429	0.795	66.008	66.008	1.0	1.0
7 d	5	median	top1	429	0.993	0.432	1.0	429	0.432	223.642	223.642	1.007	1.007
7 d	2	mean	top20	2097	0.764	0.726	0.777	2094	0.927	55.495	21.467	0.782	1.002
7 d	2	mean	top10	2097	0.764	0.697	0.78	2094	0.89	64.694	34.02	0.782	1.002
7 d	2	mean	top5	2097	0.764	0.632	0.763	2094	0.808	84.089	67.972	0.779	1.001
7 d	2	mean	top2	2097	0.764	0.572	0.758	2094	0.749	55.767	79.238	0.78	1.005
7 d	2	mean	top1	2097	0.764	0.334	0.703	2094	0.502	85.359	133.909	0.777	0.998
7 d	2	median	top20	2097	0.764	0.714	0.775	2094	0.918	47.192	12.678	0.782	1.001
7 d	2	median	top10	2097	0.764	0.678	0.78	2094	0.855	43.591	49.499	0.782	1.003
7 d	2	median	top5	2097	0.764	0.644	0.764	2094	0.842	53.267	38.116	0.78	1.002
7 d	2	median	top2	2097	0.764	0.579	0.75	2094	0.748	36.796	37.933	0.748	0.998
7 d	2	median	top1	2097	0.764	0.473	0.72	2094	0.63	196.789	220.85	0.656	0.912
14 d	5	mean	top20	432	1.0	0.979	1.0	432	0.979	25.708	25.708	0.999	0.999
14 d	5	mean	top10	432	1.0	1.0	1.0	432	1.0	5.12	5.12	1.0	1.0
14 d	5	mean	top5	432	1.0	0.909	1.0	432	0.909	64.33	64.33	1.0	1.0
14 d	5	mean	top2	432	1.0	0.8	1.0	432	0.8	123.062	123.062	0.992	0.992
14 d	5	mean	top1	432	1.0	1.0	1.0	432	1.0	1.462	1.462	1.0	1.0
14 d	5	median	top20	432	1.0	1.0	1.0	432	1.0	4.942	4.942	1.0	1.0
14 d	5	median	top10	432	1.0	1.0	1.0	432	1.0	0.796	0.796	1.0	1.0
14 d	5	median	top5	432	1.0	0.909	1.0	432	0.909	36.197	36.197	1.0	1.0
14 d	5	median	top2	432	1.0	0.795	1.0	432	0.795	155.74	155.74	1.0	1.0
14 d	5	median	top1	432	1.0	0.676	1.0	432	0.676	95.862	95.862	1.0	1.0
14 d	2	mean	top20	2405	0.885	0.847	0.886	2400	0.955	24.495	8.363	0.895	1.0
14 d	2	mean	top10	2405	0.885	0.819	0.888	2400	0.917	27.979	6.692	0.896	1.0
14 d	2	mean	top5	2405	0.885	0.766	0.859	2400	0.89	68.268	48.386	0.895	1.0

window	grid	stat	thr	C_W	C_W area	J_end	O_M	S	J_sh	D_end km	D_sh km	R_end	R_sh
14 d	2	mean	top2	2405	0.885	0.745	0.889	2400	0.844	64.951	61.875	0.909	1.002
14 d	2	mean	top1	2405	0.885	0.53	0.853	2400	0.599	96.761	163.734	0.924	0.997
14 d	2	median	top20	2405	0.885	0.828	0.886	2400	0.937	10.418	11.577	0.896	1.0
14 d	2	median	top10	2405	0.885	0.799	0.892	2400	0.894	10.791	23.55	0.895	1.0
14 d	2	median	top5	2405	0.885	0.764	0.867	2400	0.904	17.261	25.281	0.895	0.998
14 d	2	median	top2	2405	0.885	0.72	0.892	2400	0.8	37.781	16.912	0.873	0.979
14 d	2	median	top1	2405	0.885	0.611	0.906	2400	0.608	120.814	157.331	0.811	0.827
wk 1*	5	mean	top20	429	0.993	0.945	0.989	429	0.956	30.376	25.866	0.989	1.001
wk 1*	5	mean	top10	429	0.993	0.887	0.977	429	0.907	38.872	41.534	0.979	1.001
wk 1*	5	mean	top5	429	0.993	0.829	1.0	429	0.829	73.611	73.611	1.004	1.004
wk 1*	5	mean	top2	429	0.993	0.795	1.0	429	0.795	64.791	64.791	1.0	1.0
wk 1*	5	mean	top1	429	0.993	0.661	1.0	429	0.661	150.439	150.439	1.007	1.007
wk 1*	5	median	top20	429	0.993	0.942	0.989	429	0.953	58.091	35.074	0.99	1.001
wk 1*	5	median	top10	429	0.993	0.931	0.977	429	0.952	78.778	45.819	0.978	1.001
wk 1*	5	median	top5	429	0.993	0.829	1.0	429	0.829	83.354	83.354	1.004	1.004
wk 1*	5	median	top2	429	0.993	0.795	1.0	429	0.795	66.008	66.008	1.0	1.0
wk 1*	5	median	top1	429	0.993	0.432	1.0	429	0.432	223.642	223.642	1.007	1.007
wk 1*	2	mean	top20	2097	0.764	0.726	0.777	2094	0.927	55.495	21.467	0.782	1.002
wk 1*	2	mean	top10	2097	0.764	0.697	0.78	2094	0.89	64.694	34.02	0.782	1.002
wk 1*	2	mean	top5	2097	0.764	0.632	0.763	2094	0.808	84.089	67.972	0.779	1.001
wk 1*	2	mean	top2	2097	0.764	0.572	0.758	2094	0.749	55.767	79.238	0.78	1.005
wk 1*	2	mean	top1	2097	0.764	0.334	0.703	2094	0.502	85.359	133.909	0.777	0.998
wk 1*	2	median	top20	2097	0.764	0.714	0.775	2094	0.918	47.192	12.678	0.782	1.001
wk 1*	2	median	top10	2097	0.764	0.678	0.78	2094	0.855	43.591	49.499	0.782	1.003
wk 1*	2	median	top5	2097	0.764	0.644	0.764	2094	0.842	53.267	38.116	0.78	1.002
wk 1*	2	median	top2	2097	0.764	0.579	0.75	2094	0.748	36.796	37.933	0.748	0.998
wk 1*	2	median	top1	2097	0.764	0.473	0.72	2094	0.63	196.789	220.85	0.656	0.912
wk 2	5	mean	top20	428	0.99	0.944	1.0	428	0.934	66.636	56.901	0.986	0.997
wk 2	5	mean	top10	428	0.99	0.891	1.0	428	0.912	79.822	64.634	0.971	0.995
wk 2	5	mean	top5	428	0.99	0.909	1.0	428	0.909	88.439	88.439	1.0	1.0
wk 2	5	mean	top2	428	0.99	1.0	1.0	428	1.0	9.908	9.908	1.0	1.0
wk 2	5	mean	top1	428	0.99	0.669	1.0	428	0.669	150.438	150.438	0.992	0.992
wk 2	5	median	top20	428	0.99	0.922	1.0	428	0.912	60.457	68.516	0.984	0.995
wk 2	5	median	top10	428	0.99	0.932	1.0	428	0.909	44.146	31.217	0.974	0.999
wk 2	5	median	top5	428	0.99	0.911	1.0	428	0.911	86.93	86.93	0.997	0.997
wk 2	5	median	top2	428	0.99	1.0	1.0	428	1.0	8.875	8.875	1.0	1.0
wk 2	5	median	top1	428	0.99	0.669	1.0	428	0.669	110.294	110.294	0.992	0.992
wk 2	2	mean	top20	2136	0.782	0.729	0.782	2131	0.917	59.566	39.785	0.797	1.0
wk 2	2	mean	top10	2136	0.782	0.724	0.788	2131	0.918	11.431	15.54	0.794	0.998
wk 2	2	mean	top5	2136	0.782	0.647	0.764	2131	0.86	79.214	21.652	0.792	0.998
wk 2	2	mean	top2	2136	0.782	0.539	0.76	2131	0.685	236.224	115.157	0.799	1.001
wk 2	2	mean	top1	2136	0.782	0.485	0.818	2131	0.569	437.447	294.381	0.822	1.005
wk 2	2	median	top20	2136	0.782	0.719	0.78	2131	0.907	70.637	58.712	0.798	1.003
wk 2	2	median	top10	2136	0.782	0.724	0.784	2131	0.923	30.398	42.026	0.794	0.993
wk 2	2	median	top5	2136	0.782	0.634	0.748	2131	0.876	111.926	41.429	0.792	0.999
wk 2	2	median	top2	2136	0.782	0.546	0.732	2131	0.687	92.745	66.963	0.767	1.001
wk 2	2	median	top1	2136	0.782	0.587	0.844	2131	0.688	63.218	57.338	0.687	0.813
wk 3	5	mean	top20	432	1.0	0.976	1.0	432	0.976	33.269	33.269	1.0	1.0
wk 3	5	mean	top10	432	1.0	0.909	1.0	432	0.909	25.689	25.689	1.002	1.002
wk 3	5	mean	top5	432	1.0	1.0	1.0	432	1.0	8.592	8.592	1.0	1.0
wk 3	5	mean	top2	432	1.0	0.8	1.0	432	0.8	151.335	151.335	0.992	0.992
wk 3	5	mean	top1	432	1.0	0.676	1.0	432	0.676	93.303	93.303	1.0	1.0
wk 3	5	median	top20	432	1.0	0.955	1.0	432	0.955	55.263	55.263	1.0	1.0
wk 3	5	median	top10	432	1.0	0.909	1.0	432	0.909	24.072	24.072	1.002	1.002
wk 3	5	median	top5	432	1.0	0.913	1.0	432	0.913	137.668	137.668	1.004	1.004
wk 3	5	median	top2	432	1.0	0.8	1.0	432	0.8	149.157	149.157	0.992	0.992
wk 3	5	median	top1	432	1.0	1.0	1.0	432	1.0	5.31	5.31	1.0	1.0
wk 3	2	mean	top20	2214	0.809	0.746	0.793	2210	0.952	39.682	18.728	0.825	1.001
wk 3	2	mean	top10	2214	0.809	0.727	0.787	2210	0.913	65.05	18.454	0.824	0.999
wk 3	2	mean	top5	2214	0.809	0.65	0.786	2210	0.832	52.111	24.681	0.823	1.001
wk 3	2	mean	top2	2214	0.809	0.678	0.891	2210	0.763	84.883	79.905	0.837	1.005
wk 3	2	mean	top1	2214	0.809	0.724	0.891	2210	0.839	19.674	19.561	0.853	1.0
wk 3	2	median	top20	2214	0.809	0.753	0.793	2210	0.935	37.402	13.985	0.824	1.001
wk 3	2	median	top10	2214	0.809	0.72	0.783	2210	0.909	32.172	20.846	0.825	0.995
wk 3	2	median	top5	2214	0.809	0.672	0.792	2210	0.849	114.944	125.985	0.825	1.004
wk 3	2	median	top2	2214	0.809	0.654	0.892	2210	0.763	60.974	31.495	0.802	1.002
wk 3	2	median	top1	2214	0.809	0.648	0.938	2210	0.807	44.93	48.179	0.752	1.046
wk 4	5	mean	top20	427	0.989	0.945	0.979	427	0.975	30.113	19.034	0.989	1.0
wk 4	5	mean	top10	427	0.989	0.932	0.978	427	0.952	43.228	34.598	0.979	1.001
wk 4	5	mean	top5	427	0.989	0.756	0.957	427	0.83	27.285	33.971	1.002	0.997
wk 4	5	mean	top2	427	0.989	0.8	1.0	427	0.8	141.943	141.943	0.992	0.992
wk 4	5	mean	top1	427	0.989	0.668	1.0	427	0.668	297.589	297.589	1.015	1.015
wk 4	5	median	top20	427	0.989	0.946	0.979	427	0.956	48.647	57.141	0.992	1.003

window	grid	stat	thr	C_W	C_W area	J_end	O_M	S	J_sh	D_end km	D_sh km	R_end	R_sh
wk 4	5	median	top10	427	0.989	0.888	0.978	427	0.907	58.572	59.294	0.98	1.001
wk 4	5	median	top5	427	0.989	0.831	0.957	427	0.828	92.331	84.47	1.004	1.0
wk 4	5	median	top2	427	0.989	0.808	1.0	427	0.808	53.181	53.181	1.0	1.0
wk 4	5	median	top1	427	0.989	0.432	1.0	427	0.432	113.934	113.934	1.007	1.007
wk 4	2	mean	top20	2236	0.816	0.673	0.748	2227	0.925	67.28	30.59	0.836	1.001
wk 4	2	mean	top10	2236	0.816	0.632	0.73	2227	0.879	82.405	16.585	0.835	1.0
wk 4	2	mean	top5	2236	0.816	0.602	0.712	2227	0.818	73.237	42.29	0.832	1.0
wk 4	2	mean	top2	2236	0.816	0.57	0.777	2227	0.73	91.465	167.507	0.838	1.007
wk 4	2	mean	top1	2236	0.816	0.516	0.851	2227	0.587	163.732	199.184	0.855	1.005
wk 4	2	median	top20	2236	0.816	0.664	0.75	2227	0.917	73.357	37.938	0.836	1.002
wk 4	2	median	top10	2236	0.816	0.615	0.722	2227	0.879	91.58	10.222	0.835	1.0
wk 4	2	median	top5	2236	0.816	0.604	0.727	2227	0.85	80.425	30.271	0.833	1.001
wk 4	2	median	top2	2236	0.816	0.553	0.713	2227	0.799	105.156	106.189	0.804	1.005
wk 4	2	median	top1	2236	0.816	0.526	0.843	2227	0.606	123.878	92.995	0.816	0.968

S7 Quality rule QC-1 and the excluded NOAA-15 record

Rule (version 1.0, adopted post hoc). For each daily file in native order, a record's OMNI values (p1–p3) are set missing when: its fit flag is 1 or 2; its 10 preceding and 10 following records lie in one contiguous run (every time step positive and ≤ 6 s) and all have fit flag 0; and, for at least one channel, every neighbour value is finite and positive and the record's value exceeds 100 × the median of the preceding ten **and** 100 × the median of the following ten (strict inequality). Records that cannot be evaluated are kept. Identity is the source SHA-256 and the zero-based native index; numerically equal values elsewhere are unaffected. Code: `src/saa/quality.py`; tests: `tests/test_omni_qc.py` (event exclusion, a real isolated one-point fit that is kept, gradual intense passes, gaps, missing/zero neighbours, duplicate and backward timestamps, positional identity, strict threshold, deterministic replay).

Population envelopes (all 155 files; "regular": two-sided ratio min(left, right) of fit-0 records whose 20 neighbours are all fit 0; "fallback": the same for isolated fit-1/2 records; "excluded ratio": smallest ratio among excluded values):

satellite	ch	records	regular n	regular max	fallback n	fallback median	fallback max kept	excluded	excluded ratio
noaa15	p1	1222897	66122	2.75	208	3.3	9.0	1	2868.8
noaa15	p2	1222897	66122	2.25	208	0.58	1.77	1	592.63
noaa15	p3	1222897	66122	1.55	208	0.14	0.82	1	126.67
noaa18	p1	1217377	75605	2.99	214	3.11	9.01	0	
noaa18	p2	1217377	75605	2.33	214	0.59	1.53	0	
noaa18	p3	1217377	75605	1.5	214	0.12	1.05	0	
noaa19	p1	1206770	70746	2.93	231	3.71	11.85	0	
noaa19	p2	1206770	70746	2.38	231	0.55	1.45	0	
noaa19	p3	1206770	70746	1.39	231	0.14	1.56	0	
metop01	p1	1333580	69544	3.28	310	5.21	19.76	0	
metop01	p2	1333580	69544	2.27	310	0.57	1.34	0	
metop01	p3	1333580	69544	1.4	310	0.14	3.43	0	
metop03	p1	1333046	67805	3.18	279	5.0	16.19	0	
metop03	p2	1333046	67805	2.65	279	0.55	1.08	0	
metop03	p3	1333046	67805	1.66	279	0.14	0.88	0	

Parameter perturbations (one at a time around the adopted rule, plus joint ratio × neighbour changes; "same set" compares the full list of excluded records, not only its size):

variant	excluded (all)	excluded (region)	same set
adopted rule	1	1	True
ratio 5x	453	450	False
ratio 10x	30	29	False
ratio 15x	4	4	False
ratio 20x	1	1	True
ratio 30x	1	1	True
ratio 50x	1	1	True
ratio 200x	1	1	True
ratio 500x	1	1	True
ratio 1000x	1	1	True
ratio 2000x	1	1	True
ratio 3000x	0	0	False
3 neighbours per side	1	1	True
5 neighbours per side	1	1	True
7 neighbours per side	1	1	True
15 neighbours per side	1	1	True

variant	excluded (all)	excluded (region)	same set
20 neighbours per side	1	1	True
30 neighbours per side	1	1	True
max step 3 s	1	1	True
max step 10 s	1	1	True
max step 20 s	1	1	True
regular-fit neighbours >= 90%	1	1	True
regular-fit neighbours >= 75%	1	1	True
regular-fit neighbours >= 50%	1	1	True
no regular-fit neighbour requirement	32	8	False
fallback fit not required	1	1	True
trigger on p1 only	1	1	True
trigger on all channels	1	1	True
no fit conditions (pure two-sided ratio)	32	8	False
ratio 10x, 5 neighbours	126	126	False
ratio 10x, 20 neighbours	3	3	False
ratio 20x, 5 neighbours	6	6	False
ratio 20x, 20 neighbours	1	1	True
ratio 30x, 5 neighbours	1	1	True
ratio 30x, 20 neighbours	1	1	True

The excluded set is unchanged in 26 of 35 variants. It grows when the ratio falls to 15× or 10× (4 and 30 records). The three records added at 15× are isolated one-point default-slope fits at low flux on the edge of the SAA, beside lower regular-fit neighbours, and their co-located directional channels are not zero: departures of the kind the documented fallback produces elsewhere, which the ratio is set to leave alone. It also grows to 32 records (8 in the study region) when the regular-fit neighbour condition is dropped; these are mostly low-flux fallback values beside near-zero neighbours, the cases that condition is designed to leave alone.

Excluded record. Zero-based native index 13867 of the NOAA-15 file for 19 January, 2024-01-19T09:14:30.051Z (`poes_n15_20240119_proc.nc`, SHA-256 `1209804e1f2e59b49f0e55f0d5be6246e89fef9df3c8641f76d6dbf679bb0594`). Stored values (float32): p1 = 36,016, p2 = 4825.2, p3 = 646.4, whereas its 20 native neighbours (±20 s) have a median p1 of 11.7. It carries OMNI fit flag 1, while all 20 neighbours carry flag 0. Two-sided p1 ratios: 3,272 (preceding) and 2,869 (following); p3 ratios 155 and 127. Its p1/p2 and p2/p3 ratios are 7.464 and 7.464, equal to $2^{2.9} = 7.464$ for energies a factor two apart: the three channels are one value extended with the default slope, not three independent enhancements. In the same record, the co-located directional MEPED P6 proton fluxes (0° and 90° telescopes), which are at least 6,098 $cm^{-2}\ s^{-1}\ sr^{-1}$ on every neighbour, are exactly zero. It is the only record QC-1 excludes in the 155 files; it is also one of the candidates of the historical triage described at the end of this section.

Native neighbourhood (±10 records; flux in $cm^{-2}\ s^{-1}\ sr^{-1}\ MeV^{-1}$ γ1 is the fitted slope at p1; P6 0°/90° are the co-located directional MEPED proton fluxes, $cm^{-2}\ s^{-1}\ sr^{-1}$):

index	UTC	lat	lon	IFC	fit	p1	p2	p3	γ1	P6 0°	P6 90°	gap_s
13857	09:14:10.051	-11.354	-29.025	-1	0	9.507	7.197	3.892	-0.4	1.085e+04	6585	2.0
13858	09:14:12.051	-11.471	-29.052	-1	0	12.86	6.568	3.853	-1.0	1.183e+04	6098	2.0
13859	09:14:14.051	-11.589	-29.079	-1	0	11.06	6.77	4.125	-0.7	7073	7317	2.0
13860	09:14:16.051	-11.707	-29.105	-1	0	10.96	6.776	4.183	-0.7	1.134e+04	7561	2.0
13861	09:14:18.052	-11.825	-29.132	-1	0	10.96	6.776	4.183	-0.7	1.366e+04	8902	2.001
13862	09:14:20.052	-11.943	-29.159	-1	0	9.912	7.408	4.162	-0.4	1.037e+04	1.134e+04	2.0
13863	09:14:22.052	-12.06	-29.186	-1	0	11.79	7.36	4.72	-0.7	1.183e+04	7317	2.0
13864	09:14:24.052	-12.178	-29.213	-1	0	12.79	5.897	4.651	-1.1	8902	1.085e+04	2.0
13865	09:14:26.052	-12.296	-29.24	-1	0	11.99	7.348	4.605	-0.7	1.183e+04	8902	2.0
13866	09:14:28.052	-12.414	-29.267	-1	0	10.58	7.334	4.605	-0.5	1.134e+04	1.037e+04	2.0
13867	09:14:30.051	-12.532	-29.293	-1	1	3.602e+04	4825	646.4	-2.9	0	0	1.999
13868	09:14:32.051	-12.649	-29.32	0	0	12.14	8.316	4.575	-0.5	1.293e+04	9878	2.0
13869	09:14:34.051	-12.767	-29.347	-1	0	10.86	8.825	4.835	-0.3	9878	1.037e+04	2.0
13870	09:14:36.051	-12.885	-29.374	-1	0	14.34	7.878	4.737	-0.9	1.366e+04	1.134e+04	2.0
13871	09:14:38.051	-13.003	-29.401	-1	0	12.97	7.434	5.028	-0.8	1.232e+04	9390	2.0
13872	09:14:40.051	-13.121	-29.428	-1	0	13	8.41	5.069	-0.6	1.512e+04	9390	2.0
13873	09:14:42.051	-13.238	-29.455	-1	0	11.61	7.959	5.362	-0.5	1.232e+04	8902	2.0
13874	09:14:44.052	-13.356	-29.482	-1	0	13.39	6.989	5.233	-0.9	1.232e+04	7805	2.001
13875	09:14:46.052	-13.474	-29.509	-1	0	15.23	7.968	5.203	-0.9	1.293e+04	1.232e+04	2.0
13876	09:14:48.052	-13.592	-29.536	-1	0	10.28	8.878	5.172	-0.2	1.037e+04	1.134e+04	2.0
13877	09:14:50.052	-13.709	-29.563	-1	0	10.09	9.987	5.137	-0.0	1.232e+04	1.134e+04	2.0

Level-1a count rates (±5 records; counts s^{-1} ">16 MeV" … ">140 MeV" are the OMNI channels P6–P9, and P8/P9 share a telemetry word and are read out on alternate records, shown as "–"; P6 0°/90° are directional; "bias V" is the OMNI bias-voltage housekeeping field):

index	UTC	>16 MeV	>35 MeV	>70 MeV	>140 MeV	P6 0°	P6 90°	bias V	fit	p1
13862	09:14:20.052	783.75	815.75	719.88	–	42.5	46.5	685.41	0	9.912
13863	09:14:22.052	855.75	855.75	–	391.88	48.5	30.0	685.41	0	11.79
13864	09:14:24.052	815.75	783.75	719.88	–	36.5	44.5	685.41	0	12.79
13865	09:14:26.052	855.75	855.75	–	391.88	48.5	36.5	685.41	0	11.99
13866	09:14:28.052	815.75	855.75	719.88	–	46.5	42.5	516.57	0	10.58
13867	09:14:30.051	903.75	487423.75	–	391.88	0.0	0.0	516.57	1	3.602e+04
13868	09:14:32.051	903.75	903.75	751.88	–	53.0	40.5	516.57	0	12.14
13869	09:14:34.051	903.75	951.75	–	427.88	40.5	42.5	516.57	0	10.86
13870	09:14:36.051	951.75	903.75	783.88	–	56.0	46.5	516.57	0	14.34
13871	09:14:38.051	903.75	903.75	–	407.88	50.5	38.5	516.57	0	12.97
13872	09:14:40.051	951.75	951.75	815.88	–	62.0	38.5	516.57	0	13

In this record the >35 MeV rate is 487,424 s^{-1}, against 752–1,056 s^{-1} on the neighbours, while the >16 MeV rate is 904 s^{-1}, inside its neighbours' range (752–1,000 s^{-1}); the ratio is 539. Every proton above 35 MeV is also above 16 MeV, so these two rates cannot both describe the same particle population. Under the documented procedure (Machol, 2012), such a count-rate vector precludes the regular fit and leaves the one-point default-slope fit, consistent with the stored flag and slopes. Elsewhere that day the >35/>16 MeV ratio never exceeds 17.0, and the second-highest >35 MeV rate is 2,752 s^{-1}. In this record 12 of the 12 directional MEPED proton count channels and 5 of the 6 electron count channels read exactly zero; 20 level-1a variables in all are zero here and non-zero in all 20 neighbours. The bias-voltage field reads a value that occurs nowhere else that day for 16 records starting one record before the event; we report this coincidence without interpreting it. None of this identifies the cause.

Cell and footprint influence. The record lies in 5° cell (−12.5°, −27.5°) and 2° cell (−13°, −29°). Per scenario and setting ("QC-1 primary" excludes it; "unfiltered" keeps it):

scenario	grid	stat	thr	lat	lon	n	cell mean	cell median	rank	selected	centroid weight
unfiltered	5	mean	top20	-12.5	-27.5	526	77.7632	9.3378	1	True	0.0655
unfiltered	5	mean	top10	-12.5	-27.5	526	77.7632	9.3378	1	True	0.0862
unfiltered	5	mean	top5	-12.5	-27.5	526	77.7632	9.3378	1	True	0.1343
unfiltered	5	mean	top2	-12.5	-27.5	526	77.7632	9.3378	1	True	0.2616
unfiltered	5	mean	top1	-12.5	-27.5	526	77.7632	9.3378	1	True	0.4058
unfiltered	5	median	top20	-12.5	-27.5	526	77.7632	9.3378	50	True	0.0084
unfiltered	5	median	top10	-12.5	-27.5	526	77.7632	9.3378	50	False	0.0
unfiltered	5	median	top5	-12.5	-27.5	526	77.7632	9.3378	50	False	0.0
unfiltered	5	median	top2	-12.5	-27.5	526	77.7632	9.3378	50	False	0.0
unfiltered	5	median	top1	-12.5	-27.5	526	77.7632	9.3378	50	False	0.0
unfiltered	2	mean	top20	-13.0	-29.0	89	415.7915	11.2671	1	True	0.0563
unfiltered	2	mean	top10	-13.0	-29.0	89	415.7915	11.2671	1	True	0.0745
unfiltered	2	mean	top5	-13.0	-29.0	89	415.7915	11.2671	1	True	0.1149
unfiltered	2	mean	top2	-13.0	-29.0	89	415.7915	11.2671	1	True	0.2207
unfiltered	2	mean	top1	-13.0	-29.0	89	415.7915	11.2671	1	True	0.3545
unfiltered	2	median	top20	-13.0	-29.0	89	415.7915	11.2671	279	True	0.0016
unfiltered	2	median	top10	-13.0	-29.0	89	415.7915	11.2671	279	False	0.0
unfiltered	2	median	top5	-13.0	-29.0	89	415.7915	11.2671	279	False	0.0
unfiltered	2	median	top2	-13.0	-29.0	89	415.7915	11.2671	279	False	0.0
unfiltered	2	median	top1	-13.0	-29.0	89	415.7915	11.2671	279	False	0.0
QC-1 primary	5	mean	top20	-12.5	-27.5	525	9.3087	9.3378	51	True	0.0083
QC-1 primary	5	mean	top10	-12.5	-27.5	525	9.3087	9.3378	51	False	0.0
QC-1 primary	5	mean	top5	-12.5	-27.5	525	9.3087	9.3378	51	False	0.0
QC-1 primary	5	mean	top2	-12.5	-27.5	525	9.3087	9.3378	51	False	0.0
QC-1 primary	5	mean	top1	-12.5	-27.5	525	9.3087	9.3378	51	False	0.0
QC-1 primary	5	median	top20	-12.5	-27.5	525	9.3087	9.3378	50	True	0.0084
QC-1 primary	5	median	top10	-12.5	-27.5	525	9.3087	9.3378	50	False	0.0
QC-1 primary	5	median	top5	-12.5	-27.5	525	9.3087	9.3378	50	False	0.0
QC-1 primary	5	median	top2	-12.5	-27.5	525	9.3087	9.3378	50	False	0.0
QC-1 primary	5	median	top1	-12.5	-27.5	525	9.3087	9.3378	50	False	0.0
QC-1 primary	2	mean	top20	-13.0	-29.0	88	11.2395	11.2517	280	True	0.0016
QC-1 primary	2	mean	top10	-13.0	-29.0	88	11.2395	11.2517	280	False	0.0
QC-1 primary	2	mean	top5	-13.0	-29.0	88	11.2395	11.2517	280	False	0.0
QC-1 primary	2	mean	top2	-13.0	-29.0	88	11.2395	11.2517	280	False	0.0
QC-1 primary	2	mean	top1	-13.0	-29.0	88	11.2395	11.2517	280	False	0.0
QC-1 primary	2	median	top20	-13.0	-29.0	88	11.2395	11.2517	281	True	0.0016
QC-1 primary	2	median	top10	-13.0	-29.0	88	11.2395	11.2517	281	False	0.0
QC-1 primary	2	median	top5	-13.0	-29.0	88	11.2395	11.2517	281	False	0.0
QC-1 primary	2	median	top2	-13.0	-29.0	88	11.2395	11.2517	281	False	0.0
QC-1 primary	2	median	top1	-13.0	-29.0	88	11.2395	11.2517	281	False	0.0

Between the two analyses, top-10%/5° footprint membership differs for 1,015 NOAA-15 regional records (`outputs/qc_sensitivity/cp5_changed_memberships.csv`), and 8 of the 100 satellite settings change their selected cells, all of them NOAA-15. Rule-based exclusion reproduces the earlier exact-record omission exactly: every pairwise distance and every event-cell value of the QC-1 scenario equals the recorded omission evidence, and every unfiltered value equals the recorded as-is evidence (checked when the claims register is built).

Five-satellite footprints by scenario (area 10^6 km²):

scenario	satellite	grid	statistic	threshold	selected	area_Mkm2	lat	lon
QC-1 primary	noaa15	5	mean	top20	87	24.857	-20.696	-54.589
QC-1 primary	noaa15	5	mean	top10	44	12.541	-21.706	-54.914
QC-1 primary	noaa15	5	mean	top5	22	6.252	-22.571	-55.735
QC-1 primary	noaa15	5	mean	top2	9	2.554	-23.001	-54.757
QC-1 primary	noaa15	5	mean	top1	5	1.405	-24.396	-55.422
QC-1 primary	noaa15	5	median	top20	87	24.857	-20.697	-54.53
QC-1 primary	noaa15	5	median	top10	44	12.541	-21.682	-54.895
QC-1 primary	noaa15	5	median	top5	22	6.252	-22.549	-55.722
QC-1 primary	noaa15	5	median	top2	9	2.554	-22.996	-54.736
QC-1 primary	noaa15	5	median	top1	5	1.405	-24.386	-55.409
QC-1 primary	noaa15	2	mean	top20	533	24.363	-20.726	-54.476
QC-1 primary	noaa15	2	mean	top10	267	12.207	-21.456	-55.262
QC-1 primary	noaa15	2	mean	top5	134	6.103	-22.381	-55.492
QC-1 primary	noaa15	2	mean	top2	54	2.455	-22.954	-55.973
QC-1 primary	noaa15	2	mean	top1	27	1.221	-23.698	-56.266
QC-1 primary	noaa15	2	median	top20	533	24.371	-20.702	-54.431
QC-1 primary	noaa15	2	median	top10	267	12.209	-21.437	-55.174
QC-1 primary	noaa15	2	median	top5	134	6.106	-22.303	-55.403
QC-1 primary	noaa15	2	median	top2	54	2.455	-22.916	-55.514
QC-1 primary	noaa15	2	median	top1	27	1.223	-23.474	-56.071
QC-1 primary	noaa18	5	mean	top20	87	24.95	-19.948	-54.995
QC-1 primary	noaa18	5	mean	top10	44	12.614	-20.915	-55.605
QC-1 primary	noaa18	5	mean	top5	22	6.309	-21.436	-55.842
QC-1 primary	noaa18	5	mean	top2	9	2.574	-21.933	-55.261
QC-1 primary	noaa18	5	mean	top1	5	1.416	-23.419	-56.437
QC-1 primary	noaa18	5	median	top20	87	24.95	-19.934	-55.001
QC-1 primary	noaa18	5	median	top10	44	12.614	-20.898	-55.582
QC-1 primary	noaa18	5	median	top5	22	6.309	-21.428	-55.818
QC-1 primary	noaa18	5	median	top2	10	2.848	-22.433	-56.345
QC-1 primary	noaa18	5	median	top1	5	1.416	-23.424	-55.526
QC-1 primary	noaa18	2	mean	top20	515	23.672	-19.873	-54.896
QC-1 primary	noaa18	2	mean	top10	258	11.863	-20.7	-55.422
QC-1 primary	noaa18	2	mean	top5	129	5.901	-21.767	-55.491
QC-1 primary	noaa18	2	mean	top2	52	2.369	-22.645	-55.901
QC-1 primary	noaa18	2	mean	top1	26	1.184	-22.748	-55.592
QC-1 primary	noaa18	2	median	top20	515	23.675	-19.866	-54.895
QC-1 primary	noaa18	2	median	top10	258	11.859	-20.749	-55.453
QC-1 primary	noaa18	2	median	top5	131	6.006	-21.493	-55.496
QC-1 primary	noaa18	2	median	top2	52	2.368	-22.722	-55.991
QC-1 primary	noaa18	2	median	top1	29	1.321	-22.757	-56.036
QC-1 primary	noaa19	5	mean	top20	87	24.998	-19.743	-55.075
QC-1 primary	noaa19	5	mean	top10	44	12.614	-20.886	-55.479
QC-1 primary	noaa19	5	mean	top5	22	6.309	-21.433	-55.642
QC-1 primary	noaa19	5	mean	top2	9	2.574	-21.938	-55.25
QC-1 primary	noaa19	5	mean	top1	5	1.416	-23.425	-56.425
QC-1 primary	noaa19	5	median	top20	87	24.984	-19.737	-54.845
QC-1 primary	noaa19	5	median	top10	44	12.614	-20.858	-55.459
QC-1 primary	noaa19	5	median	top5	22	6.309	-21.4	-55.786
QC-1 primary	noaa19	5	median	top2	9	2.574	-21.912	-55.243
QC-1 primary	noaa19	5	median	top1	5	1.416	-23.427	-55.538
QC-1 primary	noaa19	2	mean	top20	537	24.703	-19.746	-54.923
QC-1 primary	noaa19	2	mean	top10	269	12.371	-20.635	-55.612
QC-1 primary	noaa19	2	mean	top5	135	6.183	-21.617	-55.474
QC-1 primary	noaa19	2	mean	top2	54	2.46	-22.659	-55.681
QC-1 primary	noaa19	2	mean	top1	27	1.228	-22.961	-56.352
QC-1 primary	noaa19	2	median	top20	537	24.697	-19.715	-54.979
QC-1 primary	noaa19	2	median	top10	269	12.371	-20.604	-55.713
QC-1 primary	noaa19	2	median	top5	135	6.174	-21.76	-56.317
QC-1 primary	noaa19	2	median	top2	56	2.563	-22.062	-55.42
QC-1 primary	noaa19	2	median	top1	32	1.457	-22.765	-55.843
QC-1 primary	metop01	5	mean	top20	87	24.893	-20.52	-55.094
QC-1 primary	metop01	5	mean	top10	44	12.541	-21.667	-54.951
QC-1 primary	metop01	5	mean	top5	22	6.252	-22.566	-55.778
QC-1 primary	metop01	5	mean	top2	9	2.554	-22.994	-54.749

scenario	satellite	grid	statistic	threshold	selected	area_Mkm2	lat	lon
QC-1 primary	metop01	5	mean	top1	5	1.416	-23.431	-55.458
QC-1 primary	metop01	5	median	top20	87	24.886	-20.549	-54.98
QC-1 primary	metop01	5	median	top10	44	12.568	-21.468	-55.384
QC-1 primary	metop01	5	median	top5	22	6.252	-22.557	-55.747
QC-1 primary	metop01	5	median	top2	9	2.554	-22.993	-54.738
QC-1 primary	metop01	5	median	top1	5	1.416	-23.439	-55.455
QC-1 primary	metop01	2	mean	top20	540	24.702	-20.59	-54.938
QC-1 primary	metop01	2	mean	top10	270	12.348	-21.38	-55.368
QC-1 primary	metop01	2	mean	top5	135	6.15	-22.357	-55.676
QC-1 primary	metop01	2	mean	top2	54	2.452	-23.078	-55.654
QC-1 primary	metop01	2	mean	top1	27	1.223	-23.543	-56.182
QC-1 primary	metop01	2	median	top20	540	24.725	-20.529	-54.888
QC-1 primary	metop01	2	median	top10	270	12.334	-21.482	-55.418
QC-1 primary	metop01	2	median	top5	135	6.156	-22.238	-55.486
QC-1 primary	metop01	2	median	top2	54	2.45	-23.179	-55.709
QC-1 primary	metop01	2	median	top1	27	1.226	-23.169	-55.377
QC-1 primary	metop03	5	mean	top20	87	24.893	-20.356	-55.37
QC-1 primary	metop03	5	mean	top10	44	12.568	-21.391	-55.59
QC-1 primary	metop03	5	mean	top5	22	6.309	-21.524	-55.91
QC-1 primary	metop03	5	mean	top2	9	2.554	-22.985	-56.272
QC-1 primary	metop03	5	mean	top1	5	1.416	-23.422	-56.452
QC-1 primary	metop03	5	median	top20	87	24.893	-20.348	-55.346
QC-1 primary	metop03	5	median	top10	44	12.568	-21.381	-55.558
QC-1 primary	metop03	5	median	top5	22	6.309	-21.527	-55.889
QC-1 primary	metop03	5	median	top2	10	2.839	-22.941	-57.322
QC-1 primary	metop03	5	median	top1	5	1.416	-23.417	-56.465
QC-1 primary	metop03	2	mean	top20	540	24.75	-20.332	-55.338
QC-1 primary	metop03	2	mean	top10	270	12.379	-21.087	-55.79
QC-1 primary	metop03	2	mean	top5	135	6.159	-22.149	-56.073
QC-1 primary	metop03	2	mean	top2	54	2.457	-22.844	-56.21
QC-1 primary	metop03	2	mean	top1	27	1.227	-23.118	-56.068
QC-1 primary	metop03	2	median	top20	540	24.738	-20.348	-55.331
QC-1 primary	metop03	2	median	top10	270	12.38	-21.077	-55.77
QC-1 primary	metop03	2	median	top5	135	6.158	-22.179	-56.206
QC-1 primary	metop03	2	median	top2	54	2.455	-22.901	-56.571
QC-1 primary	metop03	2	median	top1	27	1.225	-23.332	-56.748
unfiltered	noaa15	5	mean	top20	87	24.857	-20.223	-53.026
unfiltered	noaa15	5	mean	top10	44	12.548	-20.964	-52.888
unfiltered	noaa15	5	mean	top5	22	6.293	-20.938	-51.752
unfiltered	noaa15	5	mean	top2	9	2.581	-19.894	-47.808
unfiltered	noaa15	5	mean	top1	5	1.432	-19.143	-43.122
unfiltered	noaa15	5	median	top20	87	24.857	-20.697	-54.53
unfiltered	noaa15	5	median	top10	44	12.541	-21.682	-54.895
unfiltered	noaa15	5	median	top5	22	6.252	-22.549	-55.722
unfiltered	noaa15	5	median	top2	9	2.554	-22.996	-54.736
unfiltered	noaa15	5	median	top1	5	1.405	-24.386	-55.409
unfiltered	noaa15	2	mean	top20	533	24.363	-20.303	-53.082
unfiltered	noaa15	2	mean	top10	267	12.212	-20.812	-53.259
unfiltered	noaa15	2	mean	top5	134	6.103	-21.343	-52.429
unfiltered	noaa15	2	mean	top2	54	2.458	-20.73	-49.875
unfiltered	noaa15	2	mean	top1	27	1.225	-19.831	-46.494
unfiltered	noaa15	2	median	top20	533	24.371	-20.702	-54.431
unfiltered	noaa15	2	median	top10	267	12.209	-21.437	-55.174
unfiltered	noaa15	2	median	top5	134	6.106	-22.303	-55.403
unfiltered	noaa15	2	median	top2	54	2.455	-22.916	-55.514
unfiltered	noaa15	2	median	top1	27	1.223	-23.474	-56.071
unfiltered	noaa18	5	mean	top20	87	24.95	-19.948	-54.995
unfiltered	noaa18	5	mean	top10	44	12.614	-20.915	-55.605
unfiltered	noaa18	5	mean	top5	22	6.309	-21.436	-55.842
unfiltered	noaa18	5	mean	top2	9	2.574	-21.933	-55.261
unfiltered	noaa18	5	mean	top1	5	1.416	-23.419	-56.437
unfiltered	noaa18	5	median	top20	87	24.95	-19.934	-55.001
unfiltered	noaa18	5	median	top10	44	12.614	-20.898	-55.582
unfiltered	noaa18	5	median	top5	22	6.309	-21.428	-55.818
unfiltered	noaa18	5	median	top2	10	2.848	-22.433	-56.345
unfiltered	noaa18	5	median	top1	5	1.416	-23.424	-55.526
unfiltered	noaa18	2	mean	top20	515	23.672	-19.873	-54.896
unfiltered	noaa18	2	mean	top10	258	11.863	-20.7	-55.422
unfiltered	noaa18	2	mean	top5	129	5.901	-21.767	-55.491
unfiltered	noaa18	2	mean	top2	52	2.369	-22.645	-55.901
unfiltered	noaa18	2	mean	top1	26	1.184	-22.748	-55.592
unfiltered	noaa18	2	median	top20	515	23.675	-19.866	-54.895
unfiltered	noaa18	2	median	top10	258	11.859	-20.749	-55.453

scenario	satellite	grid	statistic	threshold	selected	area_Mkm2	lat	lon
unfiltered	noaa18	2	median	top5	131	6.006	-21.493	-55.496
unfiltered	noaa18	2	median	top2	52	2.368	-22.722	-55.991
unfiltered	noaa18	2	median	top1	29	1.321	-22.757	-56.036
unfiltered	noaa19	5	mean	top20	87	24.998	-19.743	-55.075
unfiltered	noaa19	5	mean	top10	44	12.614	-20.886	-55.479
unfiltered	noaa19	5	mean	top5	22	6.309	-21.433	-55.642
unfiltered	noaa19	5	mean	top2	9	2.574	-21.938	-55.25
unfiltered	noaa19	5	mean	top1	5	1.416	-23.425	-56.425
unfiltered	noaa19	5	median	top20	87	24.984	-19.737	-54.845
unfiltered	noaa19	5	median	top10	44	12.614	-20.858	-55.459
unfiltered	noaa19	5	median	top5	22	6.309	-21.4	-55.786
unfiltered	noaa19	5	median	top2	9	2.574	-21.912	-55.243
unfiltered	noaa19	5	median	top1	5	1.416	-23.427	-55.538
unfiltered	noaa19	2	mean	top20	537	24.703	-19.746	-54.923
unfiltered	noaa19	2	mean	top10	269	12.371	-20.635	-55.612
unfiltered	noaa19	2	mean	top5	135	6.183	-21.617	-55.474
unfiltered	noaa19	2	mean	top2	54	2.46	-22.659	-55.681
unfiltered	noaa19	2	mean	top1	27	1.228	-22.961	-56.352
unfiltered	noaa19	2	median	top20	537	24.697	-19.715	-54.979
unfiltered	noaa19	2	median	top10	269	12.371	-20.604	-55.713
unfiltered	noaa19	2	median	top5	135	6.174	-21.76	-56.317
unfiltered	noaa19	2	median	top2	56	2.563	-22.062	-55.42
unfiltered	noaa19	2	median	top1	32	1.457	-22.765	-55.843
unfiltered	metop01	5	mean	top20	87	24.893	-20.52	-55.094
unfiltered	metop01	5	mean	top10	44	12.541	-21.667	-54.951
unfiltered	metop01	5	mean	top5	22	6.252	-22.566	-55.778
unfiltered	metop01	5	mean	top2	9	2.554	-22.994	-54.749
unfiltered	metop01	5	mean	top1	5	1.416	-23.431	-55.458
unfiltered	metop01	5	median	top20	87	24.886	-20.549	-54.98
unfiltered	metop01	5	median	top10	44	12.568	-21.468	-55.384
unfiltered	metop01	5	median	top5	22	6.252	-22.557	-55.747
unfiltered	metop01	5	median	top2	9	2.554	-22.993	-54.738
unfiltered	metop01	5	median	top1	5	1.416	-23.439	-55.455
unfiltered	metop01	2	mean	top20	540	24.702	-20.59	-54.938
unfiltered	metop01	2	mean	top10	270	12.348	-21.38	-55.368
unfiltered	metop01	2	mean	top5	135	6.15	-22.357	-55.676
unfiltered	metop01	2	mean	top2	54	2.452	-23.078	-55.654
unfiltered	metop01	2	mean	top1	27	1.223	-23.543	-56.182
unfiltered	metop01	2	median	top20	540	24.725	-20.529	-54.888
unfiltered	metop01	2	median	top10	270	12.334	-21.482	-55.418
unfiltered	metop01	2	median	top5	135	6.156	-22.238	-55.486
unfiltered	metop01	2	median	top2	54	2.45	-23.179	-55.709
unfiltered	metop01	2	median	top1	27	1.226	-23.169	-55.377
unfiltered	metop03	5	mean	top20	87	24.893	-20.356	-55.37
unfiltered	metop03	5	mean	top10	44	12.568	-21.391	-55.59
unfiltered	metop03	5	mean	top5	22	6.309	-21.524	-55.91
unfiltered	metop03	5	mean	top2	9	2.554	-22.985	-56.272
unfiltered	metop03	5	mean	top1	5	1.416	-23.422	-56.452
unfiltered	metop03	5	median	top20	87	24.893	-20.348	-55.346
unfiltered	metop03	5	median	top10	44	12.568	-21.381	-55.558
unfiltered	metop03	5	median	top5	22	6.309	-21.527	-55.889
unfiltered	metop03	5	median	top2	10	2.839	-22.941	-57.322
unfiltered	metop03	5	median	top1	5	1.416	-23.417	-56.465
unfiltered	metop03	2	mean	top20	540	24.75	-20.332	-55.338
unfiltered	metop03	2	mean	top10	270	12.379	-21.087	-55.79
unfiltered	metop03	2	mean	top5	135	6.159	-22.149	-56.073
unfiltered	metop03	2	mean	top2	54	2.457	-22.844	-56.21
unfiltered	metop03	2	mean	top1	27	1.227	-23.118	-56.068
unfiltered	metop03	2	median	top20	540	24.738	-20.348	-55.331
unfiltered	metop03	2	median	top10	270	12.38	-21.077	-55.77
unfiltered	metop03	2	median	top5	135	6.158	-22.179	-56.206
unfiltered	metop03	2	median	top2	54	2.455	-22.901	-56.571
unfiltered	metop03	2	median	top1	27	1.225	-23.332	-56.748

Maximum pairwise centroid distance per scenario, grid, statistic and threshold:

scenario	grid_deg	statistic	threshold	satellite_a	satellite_b	distance_km
QC-1 primary	2	mean	top1	noaa15	noaa18	126.11
QC-1 primary	2	mean	top10	noaa15	noaa19	98.27
QC-1 primary	2	mean	top2	metop01	metop03	62.59
QC-1 primary	2	mean	top20	noaa15	noaa19	118.46
QC-1 primary	2	mean	top5	noaa19	metop03	85.59
QC-1 primary	2	median	top1	metop01	metop03	141.28

scenario	grid_deg	statistic	threshold	satellite_a	satellite_b	distance_km
QC-1 primary	2	median	top10	noaa15	noaa19	108.19
QC-1 primary	2	median	top2	noaa19	metop03	150.63
QC-1 primary	2	median	top20	noaa15	noaa19	123.74
QC-1 primary	2	median	top5	noaa15	noaa19	111.95
QC-1 primary	5	mean	top1	noaa15	metop03	150.61
QC-1 primary	5	mean	top10	noaa15	noaa18	113.43
QC-1 primary	5	mean	top2	noaa19	metop03	156.72
QC-1 primary	5	mean	top20	noaa15	noaa19	117.45
QC-1 primary	5	mean	top5	noaa15	noaa19	126.9
QC-1 primary	5	median	top1	noaa15	metop03	152.1
QC-1 primary	5	median	top10	noaa15	noaa18	112.53
QC-1 primary	5	median	top2	noaa15	metop03	264.78
QC-1 primary	5	median	top20	noaa15	noaa19	111.7
QC-1 primary	5	median	top5	noaa19	metop01	128.69
unfiltered	2	mean	top1	noaa15	metop01	1082.39
unfiltered	2	mean	top10	noaa15	metop03	264.6
unfiltered	2	mean	top2	noaa15	metop03	694.98
unfiltered	2	mean	top20	noaa15	metop03	235.22
unfiltered	2	mean	top5	noaa15	metop03	386.87
unfiltered	2	median	top1	metop01	metop03	141.28
unfiltered	2	median	top10	noaa15	noaa19	108.19
unfiltered	2	median	top2	noaa19	metop03	150.63
unfiltered	2	median	top20	noaa15	noaa19	123.74
unfiltered	2	median	top5	noaa15	noaa19	111.95
unfiltered	5	mean	top1	noaa15	metop03	1459.9
unfiltered	5	mean	top10	noaa15	metop03	284.09
unfiltered	5	mean	top2	noaa15	metop03	940.77
unfiltered	5	mean	top20	noaa15	metop03	244.92
unfiltered	5	mean	top5	noaa15	metop01	453.46
unfiltered	5	median	top1	noaa15	metop03	152.1
unfiltered	5	median	top10	noaa15	noaa18	112.53
unfiltered	5	median	top2	noaa15	metop03	264.78
unfiltered	5	median	top20	noaa15	noaa19	111.7
unfiltered	5	median	top5	noaa19	metop01	128.69

Historical triage census. Before QC-1 existed, a frozen two-sided screen flagged a positive value exceeding 100 times the median of each of its ten preceding and ten following native records, regardless of fit flags, where all 20 neighbours were finite and positive and no gap exceeded three cadences. Run with the original extractor over 6,313,670 native records (8,727,210 channel values) and replayed exactly here, it flagged 39 values at 32 records, with 24.5–49.9% evaluable per satellite/channel. All its candidates carry fit flag 1 or 2, and most are low-flux fallback values beside near-zero neighbours, outside the SAA. It removed nothing; QC-1 adds the regular-fit neighbour condition that separates those cases.

satellite	channel	source_records	negative	zero	evaluable_fraction	candidates
noaa15	p1	1222897	60	60432	0.473	10
noaa18	p1	1217377	169	66869	0.443	14
noaa19	p1	1206770	23	64690	0.452	5
noaa19	p2	1206770	23	65722	0.445	5
noaa19	p3	1206770	23	221229	0.245	2
metop01	p1	1333580	0	65787	0.463	3
metop03	p1	1333046	0	58245	0.499	0

S8 Descriptive magnetic context

NOAA-supplied IGRF-based quantities in the files (`Btot_sat`, `L_IGRF`) are model values, used here only to describe where footprint samples lie. The NOAA manual (2013) describes IGRF evaluation during processing (Green, 2013); the IGRF generation used for these 2024 files is not stated in the file metadata we inspected. For the principal footprint (5° mean, top 10%) of each spacecraft, 100% of footprint samples fall below that spacecraft's regional first quartile of `Btot_sat`, and 90% of footprint samples lie within the lowest 11.7–12.1% of its regional `Btot_sat` distribution. The column "separation" is a standardized median difference, $(m_{\mathrm{out}} - m_{\mathrm{in}})/(0.5(\mathrm{IQR}_{\mathrm{in}} + \mathrm{IQR}_{\mathrm{out}}))$, where m and IQR are the median and interquartile range of `Btot_sat` for regional samples inside and outside the footprint; positive values mean lower field strength inside. It ranges from 1.573 to 1.596.

satellite	footprint samples	frac < Q1(Btot)	Btot fraction for 90%	separation
noaa15	21405	1.0	0.1186	1.596

satellite	footprint samples	frac < Q1(Btot)	Btot fraction for 90%	separation
noaa18	21647	1.0	0.1209	1.5728
noaa19	20829	1.0	0.1204	1.5843
metop01	23004	1.0	0.1177	1.5726
metop03	22815	1.0	0.1166	1.5834

This is co-location with low modelled field strength, consistent with these particle-defined footprints lying in the region of weak field. It is a description of these footprints, not a test of any boundary, and it does not establish causation.

Retired material. An earlier combined classification (legacy "v1") counted low-Btot and Btot-dominance supporters separately and compared Btot against a linear magnetic-local-time separation. That MLT summary depends on the clock origin (a 12-hour shift changes a wrapping test case from 0.067 to 5.000), and "separate counts" differs from the documented "both criteria" wording. The legacy output and its dominance results are retired: they are kept in the audit archive for provenance and are not current findings. No replacement circular statistic is claimed comparable to Btot.

S9 Reproduction and verification summary

The package's `REPRODUCE.md` gives the single entry command (`python scripts/reproduce_paper.py`), the environment files and the order of steps: input hash verification; regeneration of processed tables, grids and all sensitivity families from raw files (QC-1 applied); the QC-1 ledger, population envelopes, perturbation study and unfiltered sensitivity; the level-1a count-rate evidence (hash-checked); the exact-record omission replayed on unfiltered data as provenance; the Phase 3 protocol replay; the Phase 4 temporal diagnostics; the audit census replay; the claims register; figures; the viewer payload (exported twice and compared byte for byte); tests and validators; and these documents. The final verification report lists exact commands and pass/fail counts.

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